

Alexandre Champagne-Ruel

NASA Postdoctoral Program Fellow

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Origins of Life • Astrobiology • Complex Systems

RESEARCH PROFILE

NASA Postdoctoral Fellow and physicist investigating the physical principles underlying the origin of life. My research integrates statistical mechanics, information theory, and evolutionary dynamics to understand how spatial environments drive the emergence of complexity. Currently developing quantitative frameworks for agnostic life detection and biosignature assessment.

ACADEMIC POSITIONS

NASA Postdoctoral Program Fellow
Arizona State University, Mathis Group

2025 – Current
Tempe, AZ, USA

EDUCATION

Ph.D. Physics

2025

Université de Montréal

Montréal, QC, Canada

- Thesis: *Spatiality in Prebiotic Evolution: Toward a Physics of the Emergence of Complexity*
- Advisor: Paul Charbonneau

M.Sc. Physics

2020

Université de Montréal

Montréal, QC, Canada

- Thesis: *From Game Theory to Exobiology: The Emergence of Cooperation as a Critical Phenomenon*
- Advisor: Paul Charbonneau

B.Sc. Physics

2018

Université de Montréal

Montréal, QC, Canada

B.Sc. Philosophy

2012

Université de Montréal

Montréal, QC, Canada

GRANTS & AWARDS

Only individual grants >\$5,000 are listed.

NASA Postdoctoral Program Fellowship NASA

146,496 USD

2025

Project: Mapping Molecular Complexity for Agnostic Life Detection

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2024

Globalink Research Award Mitacs Canada

6,000 CAD

2024

Google Cloud Research Grant Google

5,000 USD

2023 – 2026

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2023

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2022

Doctoral Scholarship Fonds de Recherche du Québec

70,000 CAD

2022

Excellence Award Université de Montréal

10,000 CAD

2018

Various Travel & Research Grants 15 awards

~30,000 CAD

2011 – 2025

PUBLICATIONS

Articles under Review

A. **Champagne-Ruel** and C. Mathis, “Spatial Patterning and Selection: How the Environment Shapes Molecular Complexity”, [10.48550/arXiv.2509.04547](https://arxiv.org/abs/10.48550/arXiv.2509.04547) (2025), (Submitted to PNAS Nexus).

Published Articles

S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origin of life: An integrated review. Part 1–Experimental methods and data repositories”, *Cell Reports Physical Science* **0**, [10.1016/j.xcrp.2026.103212](https://doi.org/10.1016/j.xcrp.2026.103212) (2026).

S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origins of life: An integrated review. Part 2: Theoretical methods and emerging trends”, *Cell Reports Physical Science* **0**, [10.1016/j.xcrp.2026.103211](https://doi.org/10.1016/j.xcrp.2026.103211) (2026).

S. Rajamani, C. Mathis, S. Asche, D. Boulesteix, C. Blanco, O. Markovitch, A. **Champagne-Ruel**, et al., “Better origin(s) of life research through collaboration across disciplines”, *Cell Reports Physical Science* **0**, [10.1016/j.xcrp.2026.103220](https://doi.org/10.1016/j.xcrp.2026.103220) (2026).

A. **Champagne-Ruel**, S. Zakaib-Bernier, and P. Charbonneau, “Diffusion and pattern formation in spatial games”, *Physical Review E* **110**, 014301 (2024).

A. **Champagne-Ruel** and P. Charbonneau, “A Mutation Threshold for Cooperative Takeover”, *Life* **12**, 254 (2022).

S. Gelin, A. **Champagne-Ruel**, and N. Mousseau, “Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons”, *Nature Communications* **11**, 3977 (2020).

PRESENTATIONS

Invited Talks

Spatiality in Prebiotic Evolution <i>Queens College, City University of New York</i>	2026 <i>New York, USA</i>
Physics of Complexity And Agnostic Life Detection <i>Trottier Institute for Research on Exoplanets</i>	2025 <i>Montréal, QC, Canada</i>
From Emergent Complexity to Reliable Life Detection <i>Arizona State University</i>	2024 <i>Tempe, AZ, USA</i>
Coopération, émergence et transitions <i>Université du Québec à Montréal</i>	2022 <i>Montréal, QC, Canada</i>

Contributed Talks

From Emergent Complexity to Reliable Life Detection <i>Biennial European Astrobiology Conference (BEACON)</i>	2025 <i>Reykjavik, Iceland</i>
Diffusion: An Overlooked Driver of Prebiotic Complexity <i>Astrobiology Science Conference (AbSciCon)</i>	2024 <i>Providence, RI, USA</i>
Théorie de l'information et origine de la vie <i>90e Congrès de l'ACFAS</i>	2023 <i>Montréal, QC, Canada</i>
A Mutation Threshold for Cooperative Takeover <i>Astrobiology Science Conference (AbSciCon)</i>	2022 <i>Atlanta, GA, USA</i>

Cooperation: An Emergent Universal Feature at the Dawn of Life <i>Interdisciplinary Origin of Life Meeting for Early Career Researchers</i>	2022 Montréal, QC, Canada
Mutation Favors the Emergence of Cooperation <i>Life and Space Poland</i>	2021 Virtual/Szczecin, Poland
La criticité dans un système évolutif artificiel <i>Centre de recherche en astrophysique du Québec (CRAQ)</i>	2019 Orford, QC, Canada

Posters

How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection <i>Astrobiology Science Conference (AbSciCon)</i>	2026 Madison, WI, USA
Mapping Molecular Complexity for Agnostic Life Detection <i>NASA Postdoctoral Program Virtual Symposium</i>	2025 Virtual
L'émergence de la coopération via l'évolution de réseaux informationnels <i>90e Congrès de l'ACFAS</i>	2023 Montréal, QC, Canada
Diffusion, structures spatiales et origine de la vie <i>90e Congrès de l'ACFAS</i>	2023 Montréal, QC, Canada
OoLEN - The Origin of Life Early-career Network: Building the Community Needed to Solve the Problem <i>AbSciCon</i>	2022 Atlanta, GA, USA
A Mutation Threshold for Cooperative Takeover <i>Gordon Research Seminar: Challenging Paradigms in Prebiotic Chemistry</i>	2022 Oxnard, CA, USA
A Mutation Threshold for Cooperative Takeover <i>Gordon Research Conference: Environments for the Origins of Life and Habitability</i>	2022 Oxnard, CA, USA
Les mutations favorisent la coopération en contexte évolutif <i>Centenaire, Département de Physique, Université de Montréal</i>	2021 Montréal, QC, Canada
Mutation Favors the Emergence of Cooperative Behavior <i>Molecular Origins of Life Munich</i>	2021 Virtual/Munich, Germany
Mutations Promote Cooperation in an Evolutionary Setting <i>XIXth ISSOL Conference</i>	2021 Virtual
Self-Organized Criticality : A Prelude to Avalanche Models of Solar Flares <i>Space Climate 7 Symposium</i>	2019 Orford, QC, Canada

SCHOOLS & SELECTIVE WORKSHOPS

Assembly Theory for Folded Matter <i>Santa Fe Institute</i>	2025 Santa Fe, NM, USA
Information Driven States of Matter <i>University of Rochester</i>	2024 Rochester, NY, USA

TEACHING EXPERIENCE

Mentor – Biocomputing Scholars program <i>Arizona State University</i>	2025 – 2026 Tempe, AZ, USA
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- Guiding 3 undergraduate students in developing computational chemistry simulations.

Undergraduate Internship Supervision

Université de Montréal

2022

Montréal, QC, Canada

- Supervised a summer research project on diffusion; resulted in journal publication and poster presentation.

Teaching Assistant – Introduction to Astrobiology

Université de Montréal

2021 – 2022

Montréal, QC, Canada

Tutoring – Undergraduate Level

Université de Montréal

2018 – 2022

Montréal, QC, Canada

SERVICE & OUTREACH

Scientific Leadership

Positions

Member of the Executive Board

Origin of Life Early-career Network (OoLEN)

2022 – Present

Origin of Life Digest ([link](#))

Editor

2021 – Present

Conference Organization

Interdisciplinary Origin of Life Meeting

Organizing Committee

2026

Tokyo, Japan

Frontiers in Astrobiology and Origins of Life Conference

Organizing Committee

2025

Reykjavik, Iceland

Origine de la vie : de l'astrophysique à la philosophie

Lead Organizer

2023

Montréal, QC, Canada

Interdisciplinary Origin of Life Meeting for Early Career Researchers

Lead Organizer

2022

Montréal, QC, Canada

Space Climate 7

Local Organizing Committee

2019

Orford, QC, Canada

Session Convener

AbSciCon

Assembly Theory Across Scales: From Molecules to Planetary Systems

2026

Madison, WI, USA

AbSciCon

Exploring Self-Assembly and Self-Organization in Prebiotic Complexity

2026

Madison, WI, USA

Media & Public Engagement

É. Beaudoin-Paul, “Entrevue avec Alexandre Champagne-Ruel, récipiendaire de la prestigieuse bourse en astrobiologie de la NASA”, [Quartier Libre](#) (2025).

A. Riopel, “Comment reconnaître la vie sur d'autres planètes”, [Le Devoir](#) (2023).

C. Tamura, “Cooperation and the Origin of Life”, Quantum Photonics Clubhouse Podcast (2022).

Reviewing Activities

- External Reviewer, NASA Solar System Science Program (2026)
- Journal of the Royal Society Open Science
- Journal of the Royal Society Interface
- Trends in Biochemical Sciences

Memberships

- International Society for Artificial Life
- Center for Research in Astrophysics of Québec
- Canadian Association of Physicists
- Canadian Astronomical Society
- Origin of Life Early-career Network
- International Society for the Study of the Origin of Life
- Complex Systems Society
- Scientific Society for Astrobiology (Founding Member)

TECHNICAL SKILLS

- **Languages:** Fluent in Spoken/Written French, English
- **Programming:** Python, C++, Fortran, Julia, R, LaTeX, MatLab, Assembly, Bash
- **Modeling:** Agent-based, Evolutionary Algorithms, Machine Learning, Network Theory, Game Theory
- **Development Workflow:** Linux, High Performance Computing (HPC)/SLURM, Git, Jekyll, Hugo, HTML5/CSS3